SUMMER TECHNICAL INTERNSHIP REPORT

ON

GENERATIVE AI, LLMS, AND AGENTIC SYSTEMS

Submitted in partial fulfilment of the requirements for
the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

SUBMITTED BY

NAME: AKURATHI SIVA RAMA NAGESWARA RAO

Reg. No.: 2400031115

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

KONERU LAKSHMAIAH EDUCATION FOUNDATION

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INTERNSHIP CERTIFICATE

DECLARATION BY CANDIDATE

I hereby declare that the work presented in this report entitled "**Generative AI, LLMs, and Agentic Systems**" in the fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering submitted in the Department of Computer Science and Engineering, Koneru Lakshmaiah Education Foundation, Green Fields, Vaddeswaram is an authentic record of my own work completed during the technical internship at **Swecha India**, carried out over the period from **May 25, 2026**, to **July 4, 2026**.

The matter embodied in this technical report has not been submitted previously for the award of any other degree, diploma, or institutional recognition.

(Student's Signature)

Name: AKURATHI SIVA RAMA NAGESWARA RAO

ID: 2400031115

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Special thanks go to my peer group and team members within the Swecha ecosystem who contributed during our sprint planning, interactive wireframing, the "Alter Ego" tasks, and repository deployments. Lastly, I express deep gratitude to my family for their unwavering encouragement throughout my engineering journey.

AKURATHI SIVA RAMA NAGESWARA RAO
Reg. No: 2400031115

ABSTRACT

The field of software engineering is experiencing a paradigm shift driven by the emergence of Large Language Models (LLMs) and intuitive development practices such as natural-language driven development, known as "vibecoding." This technical report chronicles an intensive internship at Swecha India, tracking the development lifecycle from raw AI theory to the deployment of complex, full-stack agentic applications.

The training layout covers the mechanics of state-of-the-art transformer architectures, tokenization parameters, and foundational semantic mappings. Leveraging advanced middleware abstractions—specifically **LangChain, LangGraph, and LangSmith—**practical projects were orchestrated to transition isolated models into integrated systems using Retrieval-Augmented Generation (RAG) and self-correcting AI Agents. The curriculum also comprehensively covered Domain Ecosystems, including Frontend and Backend integrations, API management, Containerization, and Model Context Protocols (MCP).

Beyond technical code optimization, the internship incorporated vital social-technical exercises. Prominent tasks included "Bridging the Digital Divide" field research, User Persona generation, and "Alter Ego" team dynamics. The capstone deliverable involved the architectural design of a Major Project: a Full Stack Application utilizing AI Agents—realized practically as an Interview Management System (IMS). The synthesized results demonstrate how modern generative workflows elevate software delivery speed while prioritizing localized, responsible open-source AI frameworks.

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1. INTRODUCTION

1.1 Overview of the Host Organization

Swecha India is a prominent socio-technical organization operating primarily in India, dedicated to promoting free and open-source software (FOSS), localization of digital resources, and engineering digital literacy across diverse socio-economic strata. The **Online AI Internship 2026** program is designed to democratize high-end compute technologies like machine learning, agentic workflows, and modern software engineering pipelines. By guiding students through collective project groups, Swecha merges baseline computer systems education with real-world infrastructure and open-source contribution paradigms.

1.2 Objectives of the Internship

The training layout was explicitly constructed to equip students with contemporary proficiency in Generative AI workflows, deep linguistic modeling layers, and full-stack framework engineering. The specific targets defined include:

- Understanding the underlying mechanics of Large Language Models (LLMs) and transformer architecture.
- Mastering agentic orchestrations using **LangChain, LangGraph, and LangSmith** to securely chain system prompts, handle conversational memory, and trace AI execution logic.
- Adapting to **"vibecoding"** approaches—leveraging interactive AI assistance to maximize rapid iterative application wireframing and prototyping.
- Integrating sophisticated AI Agents into full-stack Domain Ecosystems, managing Frontend, Backend, and Database containerization via APIs.
- Executing hands-on projects, transitioning from Mini Projects (School Simulations) to a robust Major Project (Full Stack Application with Agents).

2. INTERNSHIP OVERVIEW & TIMELINE

2.1 Roles and Responsibilities

As an engineering intern, tracking progress milestones mandated continuous updates across development sandboxes, resolving package alignment issues, and actively contributing to code-level review sessions. Team breakouts required translating open-ended objectives into concise user personas, wireframing features, executing Git deployments, and managing multi-model API variables. Responsibilities also extended to social impact activities, such as executing "Ground Reality Checks" and participating in "Alter Ego" team dynamics tasks.

2.2 Comprehensive Weekly Curriculum (May - July 2026)

The official schedule spanned precisely across a 6-week window from May 25, 2026, to July 4, 2026. The structured roadmap successfully completed during this period is itemized below:

Dates	Syllabus Milestones & Core Topics
May 25 - May 31 Foundations	Induction Program. Exploration of **Vibecoding and Essentials**. Deep dive into the Dataset to Model Journey and understanding 'How LLMs work'. Deconstructing Transformers, Tokens, and Embeddings.
June 1 - June 7 Mechanics & Planning	Vector DBs, Multi-models. Operational terminologies: COT, RAG, and AI Agents. Execution of the *Bridging the digital divide activity*. Teams & Project planning, Wireframing, and Story Boarding.
June 8 - June 14 Sprints & Infrastructure	Project Initiation. 1st iteration of project and Story Board Reviews. Retrospective on software development processes. Sessions on Technology, Data Centers, and Environmental Impacts. Team Breakouts and Pending Reviews.
June 15 - June 21 Git & Prompt Eng.	Developing User Personas & Industry Project Guidelines. Git Session and Cloud Deployments. Advanced **Prompt Engineering** aimed at building Mini Project 1 (School Education Simulations).

Dates	Syllabus Milestones & Core Topics
June 22 - June 28 Full Stack & Ecosystems	Final Review-2 & Presentations. Full Stack Technical Sessions leading into the implementation of **Major Project 2 - Full Stack Application with Agents**. Comprehensive overview of **Domain Ecosystems** (Frontend, Backend, Containers, APIs).
June 29 - July 4 Agentic Workflows	Advanced Architecture and Model Context Protocol (MCP). Deep integrations of **AI Agents - Agentic Workflows using Langchain, Langgraph, and Langsmith**. Final Project Wrap-up, Impact Assessments, and Capstone Evaluation.

3. TECHNICAL CORE CONCEPTS

3.1 Foundations of LLMs & Transformers

The technical onboarding prioritized a granular understanding of how models process data. Large Language Models operate by translating alphanumeric string patterns into structured tokens mapped in high-dimensional vector spaces. The fundamental architecture hinges upon the multi-head self-attention module within Transformer layers, allowing models to mathematically align weight scores relative to other contextual strings. This foundational knowledge was critical for setting up the baseline for multi-model integration.

3.2 Agentic Workflows: LangChain, LangGraph & LangSmith

Migrating beyond simple, static prompt injections required orchestrating autonomous systems. The curriculum focused heavily on the Lang-ecosystem during the final weeks:

- **LangChain:** Utilized as the primary orchestrator layer wrapping LLM components into standard operational chains, facilitating Retrieval-Augmented Generation (RAG) by interacting with Vector Databases.
- **LangGraph:** Implemented to build stateful, multi-actor applications. By modeling logic flows as graphs (nodes and edges), we enabled cyclical, self-correcting AI Agent workflows capable of repeating tasks until success conditions are met.
- **LangSmith:** Deployed as a debugging and tracing platform, allowing developers to visually track the execution steps, token usage, and latency of complex LLM calls in real-time.

3.3 Domain Ecosystems, Containers, and MCP

To deploy AI practically, the underlying application architecture must be robust. The training dedicated extensive sessions to "Domain Ecosystems," breaking down the software delivery pipeline. This involved bridging React-based Frontend interfaces with Spring Boot/Python Backend routers via robust APIs.

Furthermore, the introduction to Containerization (e.g., Docker) ensured that AI logic could be deployed consistently across environments without dependency conflicts. Finally, principles surrounding the Model Context Protocol (MCP) were explored to standardize how AI models interface dynamically with varied local and remote data sources.

4. PROJECT EXECUTION & SOCIAL IMPACT TASKS

4.1 Foundational Tasks: Digital Divide & Alter Ego

Reflecting Swecha's socio-technical philosophy, the internship required completing specific non-coding field tasks to better understand the user landscape.

- **Bridging the Digital Gap & Ground Reality Check:** Teams conducted field assessments to map technical discrepancies and connectivity stability impacting regional users, compiling data into actionable User Personas.
- **Alter Ego (Team Dynamics):** This task focused on internal team synchronization, simulating diverse perspectives to optimize collaboration during Git deployments and Agile software sprints.

4.2 Mini Project 1: School Education Simulations

Conducted during the Prompt Engineering sprint, this localized mini-project aimed to design structured semantic engines that generate real-time interactive simulation scripts for science pedagogy. Using LangChain prompt templates, we engineered simulation states where an LLM dynamically acts as a virtual laboratory guide, allowing students to alter parameters and receive step-by-step logic responses.

4.3 Major Project 2: Full Stack Agentic Application (IMS)

The capstone of the internship was the execution of the "Major Project 2 - Full Stack Application with agents". I elected to build an **Interview Management System (IMS)**, directly applying the curriculum's focus on Domain Ecosystems and AI Agents.

The IMS leverages a full-stack architecture to streamline the recruitment lifecycle. The frontend dashboard allows recruiters to manage candidate pipelines, while the backend utilizes LangChain and LangGraph workflows to autonomously parse uploaded resumes via RAG methodologies. A dedicated AI Agent is deployed within the system to cross-reference candidate skills against job descriptions and automatically generate customized, highly technical interview question sets, completely automating the preliminary screening phase.

5. SUMMARY OF OUTCOMES

5.1 Acquired Technical Competencies

The continuous technical engagement from May 25 to July 4 has significantly reinforced my engineering repertoire:

- **Agentic Engineering:** Mastered the deployment of autonomous AI agents using LangChain and LangGraph to build cyclical logic systems.
- **Full-Stack Architecture:** Gained strong proficiency in integrating AI models within complete Domain Ecosystems, managing API gateways between frontend clients and containerized backend services.
- **System Observability:** Developed skills using LangSmith to trace, evaluate, and debug complex LLM application pipelines.
- **Agile Team Collaboration:** Developed strong professional competencies handling multi-developer Git workflows, wireframing product user flows, and conducting project retrospective sprints.

5.2 Future Horizons of Full-Stack AI Integration

The paradigms explored throughout this internship demonstrate that the future of technical software engineering relies heavily on coordinating autonomous agentic teams within robust full-stack environments. The successful deployment of the Interview Management System (IMS) serves as a foundational blueprint. Moving forward, I plan to expand upon these open-source implementations by constructing localized, lightweight language instances that operate safely within containerized cloud architectures, bridging data security with high-end AI utility.

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